

- 3 Fig. 3.1 is a scanning electron micrograph of a pair of human chromosomes in a stage of the mitotic cell cycle.

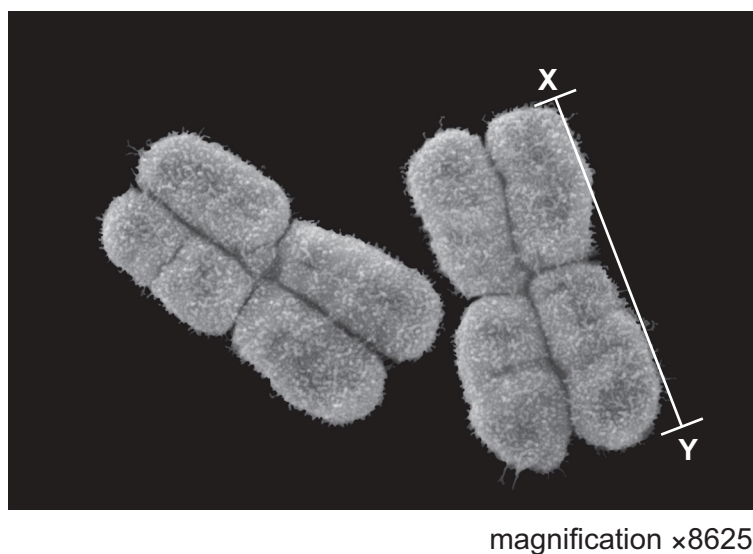


Fig. 3.1

- (a) Calculate the actual length, to the nearest $0.1\ \mu\text{m}$, of the chromosome in Fig. 3.1 indicated by the line **X–Y**.

Write the formula you used to make your calculation.

formula

actual length = μm [2]

- (b) Outline **one** feature of Fig. 3.1 that confirms the microscope used to obtain the image is a scanning electron microscope and **not** a transmission electron microscope.

.....

 [1]

- (c) With reference to Fig. 3.1, explain how it is possible to deduce that DNA replication has already occurred.

.....

 [1]

[Total: 4]